

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416343
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Donahue; Joel et al.

Shock absorber and manufacturing method of shock absorber

Abstract

In this shock absorber, a piston includes a first piston body, a second piston body, and a piston band. The first piston body includes a first passage, a second passage, a first seat, and a second seat. The first seat communicates with the first passage and on which a first damping valve is placed. A second damping valve is placed on the second seat. The second piston body includes a third passage, a fourth passage, a third seat, and a fourth seat. The third seat communicates with the fourth passage and on which a third damping valve is placed. A fourth damping valve is placed on the fourth seat.

Inventors: Donahue; Joel (Novi, MI), Truitt; Jordan (Farmington Hills, MI), Yokota; Takeru (Hitachinaka, JP), Minamino; Koji (Hitachinaka, JP)

Applicant: Hitachi Astemo, Ltd. (Ibaraki, JP)

Family ID: 1000008819729

Assignee: Hitachi Astemo, Ltd. (Ibaraki, JP)

Appl. No.: 17/954390

Filed: September 28, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240102528 A1	Mar. 28, 2024

Publication Classification

Int. Cl.: F16F9/18 (20060101); F16F9/32 (20060101); F16F9/348 (20060101); F16F9/36 (20060101); B60G13/08 (20060101)

U.S. Cl.:

CPC **F16F9/185** (20130101); **F16F9/3214** (20130101); **F16F9/3271** (20130101); **F16F9/348** (20130101); **F16F9/368** (20130101); B60G13/08 (20130101); B60G2202/24 (20130101); B60G2206/41 (20130101); B60G2800/162 (20130101); F16F2222/12 (20130101); F16F2226/04 (20130101); F16F2232/08 (20130101); F16F2234/02 (20130101)

Field of Classification Search

CPC: F16F (9/185); F16F (9/3214); F16F (9/3271); F16F (9/348); F16F (9/368); F16F (2222/12); F16F (2226/04); F16F (2232/08); B60G (13/08); B60G (2202/24); B60G (2206/41); B60G (2800/162)

USPC: 188/282.1; 188/282.2; 188/282.5; 188/282.6; 188/282.8; 188/322.15

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4895229	12/1989	Kato	188/280	F16F 9/3484
4964493	12/1989	Yamaura et al.	N/A	N/A
5042624	12/1990	Furuya	188/322.22	F16F 9/348
8800729	12/2013	Yabe	188/266.6	F16F 9/465
9182005	12/2014	Goldasz	N/A	F16F 9/3484
9777790	12/2016	Mizuno	N/A	F16F 9/3485
2009/0294232	12/2008	Masahiro	N/A	N/A
2013/0037361	12/2012	Park et al.	N/A	N/A
2014/0150897	12/2013	Ashiba	N/A	N/A
2016/0258504	12/2015	Kazuyuki et al.	N/A	N/A
2024/0102530	12/2023	Sankaran	N/A	F16F 9/22
2024/0309933	12/2023	Truitt	N/A	F16F 9/49

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
01-149037	12/1988	JP	N/A
02-113142	12/1989	JP	N/A
2009-287731	12/2008	JP	N/A
2010-107003	12/2009	JP	N/A
2014-129865	12/2013	JP	N/A
2015-086966	12/2014	JP	N/A
2019-163768	12/2018	JP	N/A
2021-181818	12/2020	JP	N/A

OTHER PUBLICATIONS

International Search Report received in corresponding International Application No. PCT/JP2023/033428 dated Dec. 5, 2023. cited by applicant

Written Opinion received in corresponding International Application No. PCT/JP2023/033428 dated Dec. 5, 2023. cited by applicant

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a shock absorber and a manufacturing method of a shock absorber.

Description of Related Art

(2) There are shock absorbers in which a piston is constituted by a plurality of piston bodies (see, for example, Japanese Unexamined Patent Application, First Publication No. 2010-107003, Japanese Unexamined Patent Application, First Publication No. 2014-129865, Japanese Unexamined Patent Application, First Publication No. H2-113142, and the specification of United States Patent Application, Publication No. 2013/0037361).

(3) There is also a shock absorber that includes a valve and a valve that opens in a region in which a piston speed is higher than that of the above-described valve (see, for example, the specification of United States Patent Application, Publication No. 2013/0037361, and Japanese Unexamined Utility Model Application, First Publication No. H1-149037).

(4) Incidentally, in shock absorbers, there has been a demand to suppress a decrease in productivity.

(5) Accordingly, an objective of the present invention is to provide a shock absorber and a manufacturing method of a shock absorber that can suppress a decrease in productivity.

SUMMARY OF THE INVENTION

(6) In order to achieve the above-described objective, one aspect of a shock absorber of the present invention includes a cylinder in which a working fluid is sealed, a piston fitted in the cylinder to be slidable and partitioning an inside of the cylinder into two chambers, and a piston rod connected to the piston and extending to an outside of the cylinder, in which the piston includes a first piston body having, a first passage and a second passage configured to allow the working fluid to flow between one chamber and the other chamber in the cylinder due to movement of the piston, a first seat communicating with the first passage and on which a first damping valve which opens at a low piston speed is placed, and a second seat having a larger diameter than the first seat, provided on the chamber side, and on which a second damping valve which opens after the first damping valve opens is placed, and a second piston body having a third passage and a fourth passage configured to allow the working fluid to flow between one chamber and the other chamber in the cylinder due to movement of the piston, a third seat communicating with the fourth passage and on which a third damping valve which opens at a low piston speed is placed, and a fourth seat having a larger diameter than the third seat, provided on the chamber side, and on which a fourth damping valve which opens after the third damping valve opens is placed, and in which the first piston body and the second piston body are connected so that the first passage communicates with the third passage and the second passage communicates with the fourth passage.

(7) One aspect of a manufacturing method of a shock absorber of the present invention is a manufacturing method of a shock absorber including a cylinder in which a working fluid is sealed, a piston fitted in the cylinder to be slidable and partitioning an inside of the cylinder into two chambers, and a piston rod connected to the piston and extending to an outside of the cylinder, in which the piston includes a first piston body having a first passage and a second passage configured to allow the working fluid to flow between one chamber and the other chamber in the cylinder due to movement of the piston, a first seat communicating with the first passage and on which a first damping valve which opens at a low piston speed is placed, and a second seat having a larger diameter than the first seat, provided on the chamber side, and on which a second damping

valve which opens after the first damping valve opens is placed, a second piston body having a third passage and a fourth passage configured to allow the working fluid to flow between one chamber and the other chamber in the cylinder due to movement of the piston, a third seat communicating with the fourth passage and on which a third damping valve which opens at a low piston speed is placed, and a fourth seat having a larger diameter than the third seat, provided on the chamber side, and on which a fourth damping valve which opens after the third damping valve opens is placed, and a piston band provided in the first piston body and the second piston body, and the manufacturing method of a shock absorber includes a first step of connecting one surface of the first piston body and one surface of the second piston body so that the first passage communicates with the third passage and the second passage communicates with the fourth passage, a second step of processing an end portion of the first passage opposite to the third passage, an end portion of the third passage opposite to the first passage, an end portion of the second passage opposite to the fourth passage, an end portion of the fourth passage opposite to the second passage, and a groove to which the piston band is attached, and a third step of attaching the piston band to the groove.

(8) According to the above-described aspects of the present invention, a decrease in productivity can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view showing a shock absorber of one embodiment according to the present invention.

(2) FIG. 2 is a cross-sectional view of a main part showing the shock absorber of one embodiment according to the present invention.

(3) FIG. 3 is a partial cross-sectional view of a main part showing the shock absorber of one embodiment according to the present invention.

(4) FIG. 4 is a partial cross-sectional view of a main part showing the shock absorber of one embodiment according to the present invention.

(5) FIG. 5 is a partial cross-sectional view of a main part showing the shock absorber of one embodiment according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(6) A shock absorber of one embodiment according to the present invention will be described on the basis of the drawings. Further, in the following description, for convenience of explanation, an upper side in the drawings will be referred to as “upper” and a lower side in the drawings will be referred to as “lower.”

(7) As shown in FIG. 1, a shock absorber **1** of the embodiment is a dual-tube type hydraulic shock absorber. The shock absorber **1** is used in suspension devices of vehicles, specifically automobiles. The shock absorber **1** includes a cylinder **2**. The cylinder **2** includes an inner tube **3** and an outer tube **4**. The inner tube **3** has a cylindrical shape. The outer tube **4** has a bottomed cylindrical shape. An inner diameter of the outer tube **4** is larger than an outer diameter of the inner tube **3**. The inner tube **3** is disposed on a radial inner side of the outer tube **4**. A central axis of the inner tube **3** and a central axis of the outer tube **4** coincide with each other. A reservoir chamber **6** is provided between the inner tube **3** and the outer tube **4**.

(8) The outer tube **4** includes a barrel part **11** and a bottom part **12**. The barrel part **11** and the bottom part **12** are formed seamlessly and integrally. The barrel part **11** has a cylindrical shape. The bottom part **12** closes a lower portion of the barrel part **11**.

(9) The shock absorber **1** includes a piston **18**. The piston **18** is inserted inside the inner tube **3** of the cylinder **2**. The piston **18** is slidably fitted in the inner tube **3** of the cylinder **2**. The piston **18** partitions the inside of the inner tube **3** into two chambers, a first chamber **19** on one side and a

second chamber **20** on the other side. The first chamber **19** is on a side opposite to the bottom part **12** with respect to the piston **18** in an axial direction of the cylinder **2**. The second chamber **20** is on a side of the bottom part **12** with respect to the piston **18** in the axial direction of the cylinder **2**. In the cylinder **2**, an oil fluid L is sealed as a working fluid in the first chamber **19** and the second chamber **20** in the inner tube **3**. In the cylinder **2**, the oil fluid L and a gas G are sealed as working fluids in the reservoir chamber **6** between the inner tube **3** and outer tube **4**.

(10) The shock absorber **1** includes a piston rod **21**. One end side of the piston rod **21** in the axial direction is disposed inside the inner tube **3** of the cylinder **2**. This one end side of the piston rod **21** is connected to the piston **18**. The other side of the piston rod **21** in the axial direction extends from the cylinder **2** to the outside of the cylinder **2**.

(11) The piston **18** is fixed to the piston rod **21**. Therefore, the piston **18** and the piston rod **21** move together. In the shock absorber **1**, a stroke in which the piston rod **21** moves in a direction to increase a protrusion amount from the cylinder **2** is an extension stroke in which the entire length increases. In the shock absorber **1**, a stroke in which the piston rod **21** moves in a direction to decrease the protrusion amount from the cylinder **2** is a compression stroke in which the entire length decreases. In the shock absorber **1**, the piston **18** moves to the first chamber **19** side during the extension stroke. In the shock absorber **1**, the piston **18** moves to the second chamber **20** side during the compression stroke.

(12) A rod guide **22** is fitted to an upper end opening side of the inner tube **3** and an upper end opening side of the outer tube **4**. A seal member **23** is fitted on an upper side of the rod guide **22** in the outer tube **4**. Both the rod guide **22** and the seal member **23** are annular. The piston rod **21** is inserted through a radial inner side of the rod guide **22** and the seal member **23**. The piston rod **21** slides in the axial direction of the rod guide **22** and the seal member **23**. The piston rod **21** extends from the inside of the cylinder **2** to the outside of the cylinder **2** from the seal member **23**.

(13) The rod guide **22** restricts movement of the piston rod **21** in the radial direction with respect to the inner tube **3** and the outer tube **4** of the cylinder **2**. The piston rod **21** is fitted in the rod guide **22** and the piston **18** is fitted in the inner tube **3**. Thereby, a central axis of the piston rod **21** and the central axis of the inner tube **3** coincide with each other. The rod guide **22** supports the piston rod **21** to be movable in the axial direction of the piston rod **21**. The seal member **23** has an outer circumferential portion in close contact with the outer tube **4**. The seal member **23** has an inner circumferential portion in close contact with an outer circumferential portion of the piston rod **21**. The piston rod **21** moves in the axial direction of the seal member **23** with respect to the seal member **23**. The seal member **23** suppresses the oil fluid L in the inner tube **3** and the high-pressure gas G and the oil fluid L in the reservoir chamber **6** leaking out to the outside.

(14) An outer circumferential portion of the rod guide **22** has a larger diameter at an upper portion than at a lower portion. The rod guide **22** is fitted to an inner circumferential portion of an upper end of the inner tube **3** at the lower portion with the small diameter. The rod guide **22** is fitted to an upper inner circumferential portion of the outer tube **4** at the upper portion having a large diameter. A base member **26** of a base valve **25** is installed on the bottom part **12** of the outer tube **4**. The base member **26** is positioned in the radial direction with respect to the outer tube **4**. An outer circumferential portion of the base member **26** has a smaller diameter at an upper portion than at a lower portion. The base member **26** is fitted to an inner circumferential portion of a lower end of the inner tube **3** at the upper portion with the small diameter. Thereby, the central axis of the inner tube **3** and the central axis of the outer tube **4** coincide with each other. The base member **26** partitions the second chamber **20** and the reservoir chamber **6**.

(15) An upper end portion of the outer tube **4** is swaged inward in the radial direction of the outer tube **4**. The seal member **23** is fixed to the cylinder **2** by being sandwiched between the swaged portion and the rod guide **22**.

(16) The piston rod **21** includes a main shaft part **27** and a mounting shaft part **28**. Both the main shaft part **27** and the mounting shaft part **28** have a rod shape.

(17) The mounting shaft part **28** has an outer diameter smaller than an outer diameter of the main shaft part **27**. The mounting shaft part **28** is disposed inside the cylinder **2**. The piston **18** is attached to the mounting shaft part **28**. The main shaft part **27** includes a shaft step part **29**. The shaft step part **29** is provided at an end portion of the main shaft part **27** on the mounting shaft part **28** side in the axial direction. The shaft step part **29** extends in a direction perpendicular to the central axis of the piston rod **21**. The mounting shaft part **28** includes a screw shaft part **31** formed at an end portion of the mounting shaft part **28** on a side opposite to the main shaft part **27** in the axial direction. The screw shaft part **31** has a male screw formed on an outer circumferential portion. A portion of the mounting shaft part **28** formed between the main shaft part **27** and the screw shaft part **31** in the axial direction of the mounting shaft part **28** is a fitting shaft part **32**. An outer circumferential surface of the fitting shaft part **32** is a cylindrical surface. The piston **18** is fitted to the fitting shaft part **32**.

(18) In the shock absorber **1**, a portion of the piston rod **21** protruding from the cylinder **2** is disposed at an upper portion to be connected to a vehicle body of a vehicle. At the same time, in the shock absorber **1**, a mounting bracket **33** fixed to the barrel part **11** of the outer tube **4** is disposed at a lower portion to be connected to a wheel side of the vehicle. The shock absorber **1** may be of a single-tube type instead of a dual-tube type. A single-tube type has a structure in which the outer tube **4** and the base valve **25** are eliminated from the shock absorber **1**, a bottomed cylindrical cylinder is provided instead of the inner tube **3**, and a free piston is provided between the bottom part of the cylinder and the piston **18**. Then, a gas is sealed between the bottom part of the cylinder and the free piston. When such a single-tube type shock absorber is used, it is also possible to connect the cylinder side to the vehicle body and the piston rod **21** to the wheel side.

(19) As shown in FIG. **2**, the piston **18** includes a first piston body **36**, a second piston body **37**, and a piston band **38**. Both the first piston body **36** and the second piston body **37** are integrally formed product made of a metal. Both the first piston body **36** and the second piston body **37** are annular. The first piston body **36** and the second piston body **37** of the piston **18** are connected to the mounting shaft part **28** of the piston rod **21**. At that time, the first piston body **36** is disposed on a side opposite to the main shaft part **27** with respect to the second piston body **37** in the axial direction of the piston rod **21**. Also, at that time, the second piston body **37** is disposed between the first piston body **36** and the shaft step part **29**.

(20) The first piston body **36** has a joint surface **41** formed on the second piston body **37** side in the axial direction. The joint surface **41** has a planar shape that extends perpendicular to a central axis of the first piston body **36**.

(21) A through hole **42** is formed at a center of the first piston body **36** in the radial direction. The through hole **42** penetrates the first piston body **36** in the axial direction of the first piston body **36**. The mounting shaft part **28** of the piston rod **21** is fitted into the through hole **42** of the first piston body **36**.

(22) The first piston body **36** includes a main body part **43**, an inner seat **45**, a first seat **46**, and a second seat **47**.

(23) The main body part **43** has an annular shape. An inner side of the main body part **43** in the radial direction is a part of the through hole **42**.

(24) The inner seat **45** has an annular shape. The inner seat **45** protrudes from the main body part **43** to a side of the first piston body **36** opposite to the joint surface **41** in the axial direction. An inner side of the inner seat **45** in the radial direction is a part of the through hole **42**.

(25) The first seat **46** has an annular shape. The first seat **46** protrudes from the main body part **43** to a side of the first piston body **36** opposite to the joint surface **41** in the axial direction. The first seat **46** has a larger diameter than the inner seat **45**. The first seat **46** is provided on a side outward from the inner seat **45** in the radial direction of the first piston body **36** to surround the inner seat **45**. A distal end surface of the first seat **46** is provided on a side opposite to the joint surface **41** with respect to a distal end surface of the inner seat **45** in the axial direction of the first piston body

36.

(26) The first piston body **36** has a passage groove **51** formed between the inner seat **45** and the first seat **46** in the radial direction of the first piston body **36**. The passage groove **51** has an annular shape extending in a circumferential direction of the first piston body **36**.

(27) A passage hole **52** is formed in the main body part **43** of the first piston body **36** between the inner seat **45** and the first seat **46** in the radial direction of the first piston body **36**. The passage hole **52** is disposed on the first seat **46** side with respect to the inner seat **45** in the radial direction of the first piston body **36**. The passage hole **52** has one end that opens to the joint surface **41** and the other end that opens to the passage groove **51**. A plurality of passage holes **52** having the same shape are provided in the first piston body **36** at regular intervals in the circumferential direction of the first piston body **36**.

(28) The second seat **47** has an annular shape. The second seat **47** protrudes from the main body part **43** to a side of the first piston body **36** opposite to the joint surface **41** in the axial direction. The second seat **47** has a larger diameter than the first seat **46**. The second seat **47** is provided on a side outward from the first seat **46** in the radial direction of the first piston body **36** to surround the first seat **46**. The second seat **47** is provided on a side opposite to the joint surface **41** with respect to the first seat **46** in the axial direction of the first piston body **36**.

(29) The first piston body **36** has an outer circumferential surface **61** formed at an outer end in the radial direction. The outer circumferential surface **61** has a cylindrical surface shape centered on the central axis of the first piston body **36**. The outer circumferential surface **61** is provided on the main body part **43**.

(30) The first piston body **36** has an opening groove part **62** on an outer circumferential portion on a radially outer side thereof. The opening groove part **62** is recessed inward in the radial direction of the first piston body **36** from the outer circumferential surface **61**. The opening groove part **62** has an annular shape centered on the central axis of the first piston body **36**. The opening groove part **62** is provided between a center of the first piston body **36** in the axial direction and the second seat **47**. A bottom surface of the opening groove part **62** overlaps the second seat **47** in position in the radial direction of the first piston body **36**.

(31) A passage hole **65** is formed in the main body part **43** of the first piston body **36**. The passage hole **65** has one end that opens to the joint surface **41** and the other end that opens to the opening groove part **62**. A plurality of passage holes **65** having the same shape are provided in the first piston body **36** at regular intervals in the circumferential direction of the first piston body **36**. In the first piston body **36**, the passage holes **52** and the passage holes **65** are alternately provided at equal pitches in the circumferential direction of the first piston body **36**. Therefore, the first piston body **36** includes the passage holes **52** and the passage holes **65** that are equal in number.

(32) The main body part **43** has a recessed part **72** that is recessed from the joint surface **41** to the inner seat **45** side in the axial direction of the first piston body **36**.

(33) In the first piston body **36**, passages in the plurality of passage holes **52** and a passage in the passage groove **51** form a first passage **75**. The first passage **75** includes a space portion between the inner seat **45** and the first seat **46**. Therefore, the first seat **46** communicates with the first passage **75**. The first passage **75** includes a space portion between the first seat **46** and the second seat **47**. Therefore, the second seat **47** also communicates with the first passage **75**.

(34) In the first piston body **36**, passages in the plurality of passage holes **65** and a passage in the opening groove part **62** form a second passage **76**.

(35) The second piston body **37** has a joint surface **81** formed on the first piston body **36** side in the axial direction. The joint surface **81** has a planar shape that extends perpendicular to a central axis of the second piston body **37**.

(36) A through hole **82** is formed at a center of the second piston body **37** in the radial direction. The through hole **82** penetrates the second piston body **37** in the axial direction of the second piston body **37**. The mounting shaft part **28** of the piston rod **21** is fitted into the through hole **82** of the

second piston body **37**.

(37) The second piston body **37** includes a main body part **83**, an inner seat **85**, a third seat **86**, a fourth seat **87**, and a protruding part **89**.

(38) The main body part **83** has an annular shape. An inner side of the main body part **83** in the radial direction is a part of the through hole **82**.

(39) The inner seat **85** has an annular shape. The inner seat **85** protrudes from the main body part **83** to a side of the second piston body **37** opposite to the joint surface **81** in the axial direction. An inner side of the inner seat **85** in the radial direction is a part of the through hole **82**.

(40) The third seat **86** has an annular shape. The third seat **86** protrudes from the main body part **83** to a side of the second piston body **37** opposite to the joint surface **81** in the axial direction. The third seat **86** has a larger diameter than the inner seat **85**. The third seat **86** is provided on a side outward from the inner seat **85** in the radial direction of the second piston body **37** to surround the inner seat **85**. A distal end surface of the third seat **86** is provided on a side opposite to the joint surface **81** with respect to a distal end surface of the inner seat **85** in the axial direction of the second piston body **37**.

(41) The second piston body **37** has a passage groove **91** formed between the inner seat **85** and the third seat **86** in the radial direction of the second piston body **37**. The passage groove **91** has an annular shape extending in the circumferential direction of the second piston body **37**.

(42) A passage hole **92** is formed in the main body part **83** of the second piston body **37** between the inner seat **85** and the third seat **86** in the radial direction of the second piston body **37**. The passage hole **92** is disposed on the third seat **86** side with respect to the inner seat **85** in the radial direction of the second piston body **37**. The passage hole **92** has one end that opens to the joint surface **81** and the other end that opens to the passage groove **91**. A plurality of passage holes **92** having the same shape are provided in the second piston body **37** at regular intervals in the circumferential direction of the second piston body **37**. The number of passage holes **92** is the same as the number of passage holes **52** of the first piston body **36**.

(43) The fourth seat **87** has an annular shape. The fourth seat **87** protrudes from the main body part **83** to a side of the second piston body **37** opposite to the joint surface **81** in the axial direction. The fourth seat **87** has a larger diameter than the third seat **86**. The fourth seat **87** is provided on a side outward from the third seat **86** in the radial direction of the second piston body **37** to surround the third seat **86**. The fourth seat **87** is provided on a side opposite to the joint surface **81** with respect to the third seat **86** in the axial direction of the second piston body **37**.

(44) The second piston body **37** has an outer circumferential surface **101** formed at an outer end in the radial direction. The outer circumferential surface **101** has a cylindrical surface shape centered on the central axis of the second piston body **37**. The outer circumferential surface **101** is provided on the main body part **83**. An outer diameter of the outer circumferential surface **101**, that is, an outer diameter of the second piston body **37** is equivalent to an outer diameter of the outer circumferential surface **61** of the first piston body **36**, that is, an outer diameter of the first piston body **36**.

(45) The second piston body **37** has an opening groove part **102** on an outer circumferential portion on a radially outer side thereof. The opening groove part **102** is recessed inward in the radial direction of the second piston body **37** from the outer circumferential surface **101**. The opening groove part **102** has an annular shape centered on the central axis of the second piston body **37**. The opening groove part **102** is provided between a center of the second piston body **37** in the axial direction and the fourth seat **87**. A bottom surface of the opening groove part **102** overlaps the fourth seat **87** in position in the radial direction of the second piston body **37**.

(46) A passage hole **105** is formed in the main body part **83** of the second piston body **37**. The passage hole **105** has one end that opens to the joint surface **81** and the other end that opens to the opening groove part **102**. A plurality of passage holes **105** having the same shape are provided in the second piston body **37** at regular intervals in the circumferential direction of the second piston

body **37**. In the second piston body **37**, the passage holes **92** and the passage holes **105** are alternately provided at equal pitches in the circumferential direction of the second piston body **37**. Therefore, the second piston body **37** includes the passage holes **92** and the passage holes **105** that are equal in number. The number of passage holes **105** is the same as the number of passage holes **65** of the first piston body **36**.

(47) The protruding part **89** protrudes from the joint surface **81** to a side opposite to the inner seat **85** in the axial direction of the second piston body **37**.

(48) In the second piston body **37**, passages in the plurality of passage holes **105** and a passage in the opening groove part **102** form a third passage **106**.

(49) In the second piston body **37**, passages in the plurality of passage holes **92** and a passage in the passage groove **91** form a fourth passage **107**. The fourth passage **107** includes a space portion between the inner seat **85** and the third seat **86**. Therefore, the third seat **86** communicates with the fourth passage **107**. The fourth passage **107** includes a space portion between the third seat **86** and the fourth seat **87**. Therefore, the fourth seat **87** also communicates with the fourth passage **107**.

(50) The first seat **46** and the third seat **86** have the same shape. The second seat **47** and the fourth seat **87** have the same shape. The passage groove **51** and the passage groove **91** have the same shape. The passage holes **52** and the passage holes **92** have the same shape. The opening groove part **62** and the opening groove part **102** have the same shape. The passage holes **65** and the passage holes **105** have the same shape.

(51) In the first piston body **36** and the second piston body **37**, the protruding part **89** of the second piston body **37** is fitted to the recessed part **72** of the first piston body **36**. As a result, the joint surface **41** of the first piston body **36** and the joint surface **81** of the second piston body **37** are in surface contact with each other. In this state, the central axis of the first piston body **36** and the central axis of the second piston body **37** coincide with each other.

(52) Also, in this state, the passage holes **52** of the first piston body **36** overlap the passage holes **105** of the second piston body **37** in position in the circumferential direction and radial direction of the first piston body **36** and the second piston body **37**. On the other hand, the passage holes **52** of the first piston body **36** do not overlap the passage holes **92** in position in the circumferential direction and radial direction of the first piston body **36** and the second piston body **37**. In other words, in this state, the first passage **75** of the first piston body **36** communicates with the third passage **106** of the second piston body **37** but does not communicate with the fourth passage **107**.

(53) Also, in this state, the passage holes **65** of the first piston body **36** overlap the passage holes **92** of the second piston body **37** in the circumferential direction and radial direction of the first piston body **36** and the second piston body **37**. On the other hand, the passage holes **65** of the first piston body **36** do not overlap the passage holes **105** in position in the circumferential direction and radial direction of the first piston body **36** and the second piston body **37**. In other words, in this state, the second passage **76** of the first piston body **36** communicates with the fourth passage **107** of the second piston body **37** but does not communicate with the third passage **106**.

(54) In this way, positions of the first piston body **36** and the second piston body **37** are aligned when the protruding part **89** is fitted to the recessed part **72**. Positions of the first piston body **36** and the second piston body **37** are not aligned in a connection state other than that. In other words, in a state other than that, the first piston body **36** and the second piston body **37** are not in a state in which the first passage **75** communicates with the third passage **106**, the second passage **76** communicates with the fourth passage **107**, and the joint surface **41** and the joint surface **81** are in surface contact with each other as described above.

(55) The protruding part **89** and the recessed part **72** form a positioning part **121**. The positioning part **121** is provided on the joint surfaces **41** and **81** at which the first piston body **36** and the second piston body **37** are connected to each other. The positioning part **121** aligns positions of the first passage **75** and the third passage **106** and aligns positions of the second passage **76** and the fourth passage **107** in the circumferential direction and radial direction of the first piston body **36**

and the second piston body **37**.

(56) The first piston body **36** and the second piston body **37** are integrated with each other in a fixed state when the protruding part **89** is press-fitted into the recessed part **72**. At that time, the first piston body **36** and the second piston body **37** are positioned in the circumferential direction and radial direction by the positioning part **121** so that the first passage **75** and the third passage **106** are communicated with each other, the second passage **76** and the fourth passage **107** are communicated with each other, and the joint surface **41** and the joint surface **81** are brought into surface contact with each other. In other words, the first piston body **36** and the second piston body **37** are connected so that the first passage **75** communicates with the third passage **106**, and the second passage **76** communicates with the fourth passage **107**.

(57) The first piston body **36**, in a state of being integrated with the second piston body **37** as described above, is formed by processing of cutting the inner seat **45**, the first seat **46**, and the second seat **47**. In other words, the first piston body **36** is formed by processing of cutting an end portion of the first passage **75** opposite to the third passage **106**.

(58) Also, the second piston body **37**, in a state of being integrated with the first piston body **36**, is formed by processing of cutting the inner seat **85**, the third seat **86**, and the fourth seat **87**. In other words, the second piston body **37** is formed by processing of cutting an end portion of the fourth passage **107** opposite to the second passage **76**.

(59) After the processing of these, the first seat **46** and the third seat **86** are mirror-symmetrical. Also, after the processing of these, the second seat **47** and the fourth seat **87** are mirror-symmetrical. Also, after the processing of these, a height difference between the inner seat **45** and the first seat **46** is larger than a height difference between the inner seat **85** and the third seat **86** by a predetermined amount.

(60) Also, the outer circumferential portions of the first piston body **36** and the second piston body **37** in the integrated state are cut at one time. At that time, the opening groove part **62** of the first piston body **36** is formed and the opening groove part **102** of the second piston body **37** is formed. In other words, the first piston body **36** is formed by processing of cutting an end portion of the second passage **76** opposite to the fourth passage **107**. The second piston body **37** is formed by processing of cutting an end portion of the third passage **106** opposite to the first passage **75**. In addition, a deep groove part **131** (groove) shown in FIG. **3** is formed in the outer circumferential portion of the first piston body **36** by cutting processing. At the same time, the outer circumferential surfaces **61** and **101** and a plurality of (specifically, four) shallow groove parts **132** (grooves) are formed on the outer circumferential portions of the first piston body **36** and the outer circumferential portion of the second piston body **37** by cutting processing. The outer circumferential surfaces **61** and **101** are formed to have the same cylindrical surface.

(61) The deep groove part **131** is recessed inward in the radial direction of the first piston body **36** from the outer circumferential surface **101**. The deep groove part **131** has an annular shape centered on the central axis of the first piston body **36**. The deep groove part **131** is formed at a portion on the second piston body **37** side with respect to the opening groove part **62** of the first piston body **36** in the axial direction.

(62) The plurality of shallow groove parts **132** have the same shape and are recessed inward in the radial direction of the first piston body **36** and the second piston body **37** from the outer circumferential surfaces **61** and **101**. The plurality of shallow groove parts **132** all have an annular shape centered on the central axes of the first piston body **36** and the second piston body **37**.

Depths of the plurality of shallow groove parts **132** from the outer circumferential surfaces **61** and **101** are smaller than a depth of the deep groove part **131** from the outer circumferential surfaces **61** and **101**. In other words, the depth of the deep groove part **131** from the outer circumferential surfaces **61** and **101** is larger than the depths of the plurality of shallow groove parts **132** from the outer circumferential surfaces **61** and **101**. In other words, a diameter of a bottom part of the deep groove part **131** is smaller than diameters of bottom parts of the plurality of shallow groove parts

132.

(63) The plurality of shallow groove parts **132** are formed in a portion of the first piston body **36** on the second piston body **37** side with respect to the deep groove part **131** in the axial direction, and a portion of the second piston body **37** on the first piston body **36** side with respect to the opening groove part **102**. In other words, the deep groove part **131** is on the second seat **47** side shown in FIG. **4** with respect to the plurality of shallow groove parts **132** in the axial direction of the first piston body **36** and the second piston body **37**. As shown in FIG. **3**, the plurality of shallow groove parts **132** are disposed to be aligned at regular intervals in the axial direction of the first piston body **36** and the second piston body **37**.

(64) The piston band **38** is made of a synthetic resin and formed in an endless strip shape. The piston band **38** includes a base body part **141**, a deep groove fitting part **142**, and a plurality of (specifically, four) shallow groove fitting parts **143**.

(65) The base body part **141** has an endless strip shape with a certain thickness.

(66) The deep groove fitting part **142** protrudes inward in the radial direction of the base body part **141** from a cylindrical surface-shaped inner circumferential surface **151** of the base body part **141**. The deep groove fitting part **142** is provided at one end of the base body part **141** in the axial direction. The deep groove fitting part **142** has an endless annular shape that continues over the entire circumference of the base body part **141**.

(67) The plurality of shallow groove fitting parts **143** have the same shape and protrude inward in the radial direction of the base body part **141** from the inner circumferential surface **151** of the base body part **141**. The plurality of shallow groove fitting parts **143** are provided at an intermediate position in the axial direction of the base body part **141**. The plurality of shallow groove fitting parts **143** all have an endless annular shape that is continuous over the entire circumference of the base body part **141**. The plurality of shallow groove fitting parts **143** are disposed to be aligned at regular intervals in the axial direction of the base body part **141**. Protrusion heights of the plurality of shallow groove fitting parts **143** from the inner circumferential surface **151** of the base body part **141** is smaller than a protrusion height of the deep groove fitting part **142** from the inner circumferential surface **151**. In other words, in the piston band **38**, inner diameters of the shallow groove fitting parts **143** are larger than an inner diameter of the deep groove fitting part **142**.

(68) The piston band **38** is attached to cover the outer circumferential portions of the first piston body **36** and the second piston body **37** integrated by press-fitting as described above. At that time, the piston band **38**, in a state of being elastically deformed to increase the diameter, covers the first piston body **36** and the second piston body **37** that is shown in FIG. **3** from the second seat **47** side shown in FIG. **4** with a side opposite to the deep groove fitting part **142** in the axial direction as the front. Then, in the piston band **38**, the deep groove fitting part **142** is first fitted to the deep groove part **131** of the first piston body **36** when the increased diameter state is released, and then the plurality of shallow groove fitting parts **143** are fitted to the plurality of shallow groove parts **132** to have a one-to-one correspondence when the increased diameter state is released. Then, the inner circumferential surface **151** of the base body part **141** comes into surface contact with the outer circumferential surfaces **61** and **101** of the first piston body **36** and the second piston body **37** to be in close contact.

(69) In this manner, the piston band **38** is integrally attached to the outer circumferential surfaces **61** and **101** of the first piston body **36** and the second piston body **37** to form the piston **18**. The piston **18** is constituted by only three parts of the first piston body **36**, the second piston body **37**, and the piston band **38**.

(70) The piston **18** is formed to include a first step, a second step, and a third step. As shown in FIG. **2**, the first step is a step of connecting the joint surface **41** that is one side of the first piston body **36** and the joint surface **81** that is one side of the second piston body **37** so that the first passage **75** communicates with the third passage **106**, and the second passage **76** communicates with the fourth passage **107**. The second step is a step of processing an end portion of the first

passage 75 opposite to the third passage 106, an end portion of the third passage 106 opposite to the first passage 75, an end portion of the second passage 76 opposite to the fourth passage 107, an end portion of the fourth passage 107 opposite to the second passage 76, and the deep groove part 131 and the plurality of shallow groove parts 132 to which the piston band 38 is attached as shown in FIG. 3. The third step is a step of attaching the piston band 38 to the deep groove part 131 and the plurality of shallow groove parts 132. In the third step, the piston band 38 is attached to the deep groove part 131 via the second seat 47 shown in FIG. 2, and then is attached to the plurality of shallow groove parts 132.

(71) The piston 18 is fitted to the fitting shaft part 32 of the piston rod 21 such that the second piston body 37 is disposed between the first piston body 36 and the shaft step part 29 in the axial direction of the piston rod 21. At that time, the piston 18 is positioned in the radial direction with respect to the piston rod 21 by fitting the through hole 82 of the second piston body 37 and the through hole 42 of the first piston body 36 onto the fitting shaft part 32.

(72) In the piston 18 in a state of being connected to the piston rod 21, the first seat 46 and the second seat 47 of the first piston body 36 are disposed on the second chamber 20 side, and the third seat 86 and the fourth seat 87 of the second piston body 37 are disposed on the first chamber 19 side. The piston 18 slides with respect to the inner tube 3 while the outer circumferential surface of the piston band 38 is in contact with the inner circumferential surface of the inner tube 3.

(73) As shown in FIG. 4, one disc 161, one disc 162, one disc 163, one disc 164, one disc 165, one disc 166, a plurality of (specifically, six) discs 167, a plurality of (specifically, three) discs 168, one disc 169, one disc 170, and one annular member 171 are stacked on the inner seat 45 of the piston 18 in order from the inner seat 45 side in the axial direction of the piston 18. The number of discs shown here is one embodiment and can be appropriately adjusted according to desired damping force characteristics.

(74) The discs 161 to 170 and the annular member 171 are all made of a metal. The discs 161 to 170 are all formed in a bored circular flat plate shape each having a certain thickness. The annular member 171 has an annular shape. The fitting shaft part 32 of the piston rod 21 is fitted to an inner side of all the discs 161 to 170 and the annular member 171.

(75) The disc 161 has an outer diameter larger than an outer diameter of the inner seat 45 of the piston 18 and smaller than an inner diameter of the first seat 46.

(76) The disc 162 has an outer diameter larger than an outer diameter of the first seat 46 of the piston 18 and smaller than an inner diameter of the second seat 47 of the piston 18. The disc 162 is separated from and comes in contact with the first seat 46 to open and close the first passage 75 at a position of the first seat 46. The disc 162 has a passage hole 182 formed at a position between the inner seat 45 of the piston 18 and an outer circumferential portion of the disc 161. The passage hole 182 penetrates the disc 162 in the axial direction of the disc 162. The passage hole 182 is an arcuate elongated hole extending in the circumferential direction of the disc 162. The passage hole 182 is opened and closed by the disc 161.

(77) The disc 163 has an outer diameter equal to the outer diameter of the disc 162. A notch part 183 is formed in an outer circumferential portion of the disc 163. The notch part 183 penetrates the disc 163 in the axial direction of the disc 163. The notch part 183 opens to the outer circumferential portion of the disc 163. A plurality of notch parts 183 are formed in the disc 163 at regular intervals in the circumferential direction of the disc 163. Inner portions of the plurality of notch parts 183 in the radial direction of the disc 163 communicate with the passage hole 182 of the disc 162. The plurality of notch parts 183 and the passage hole 182 constitute a fixed orifice 184. The fixed orifice 184 is opened and closed by the disc 161. The discs 161 and 162 constitute a check valve 185. The check valve 185 allows a flow of the oil fluid L from the fixed orifice 184 to the passage in the passage groove 51 and restricts a flow of the oil fluid L from the passage in the passage groove 51 to the fixed orifice 184.

(78) The disc 164 has an outer diameter equal to the outer diameter of the disc 163. The disc 164

covers the plurality of notch parts **183** of the disc **163** from a side opposite to the disc **162** to form the fixed orifice **184**. Therefore, the fixed orifice **184** and the check valve **185** that allows the oil fluid L to flow in only one direction to the fixed orifice **184** are formed by a plurality of laminated discs **161** to **164**.

(79) The disc **165** has an outer diameter smaller than the outer diameter of the disc **164** and slightly larger than the outer diameter of the inner seat **45**.

(80) The disc **166** has an outer diameter equal to an outer diameter of the second seat **47** of the piston **18**. The disc **166** is separated from and comes in contact with the second seat **47** to open and close the first passage **75** at a position of the second seat **47**. A notch part **186** is formed in an outer circumferential portion of the disc **166**. The notch part **186** penetrates the disc **166** in the axial direction of the disc **166**. The notch part **186** opens to the outer circumferential portion of the disc **166**. A plurality of notch parts **186** are formed in the disc **166** at regular intervals in the circumferential direction of the disc **166**. The plurality of notch parts **186** extend to cross the second seat **47** in the radial direction. The plurality of notch parts **186** form a fixed orifice **187** in the first passage **75** with the disc **166** in contact with the second seat **47**. A flow path cross-sectional area of the fixed orifice **187** is larger than a flow path cross-sectional area of the fixed orifice **184**.

(81) The plurality of discs **167** have the same outer diameter, and outer diameters thereof are each the same as the outer diameter of the disc **166**. Of the plurality of discs **167**, the disc **167** on a side closest to the disc **166** covers the plurality of notch parts **186** of the disc **166** from a side opposite to the second seat **47** to form the fixed orifice **187**.

(82) The plurality of discs **168** have the same outer diameter, and outer diameters thereof are each smaller than the outer diameter of the disc **167** and larger than the outer diameter of the disc **165**.

(83) The disc **169** has an outer diameter smaller than the outer diameter of the disc **168** and equal to the outer diameter of the disc **165**.

(84) The disc **170** has an outer diameter larger than the outer diameter of the disc **169** and equal to the outer diameter of the disc **167**.

(85) The annular member **171** has an outer diameter smaller than the outer diameter of the disc **170** and larger than the outer diameter of the disc **169**. The annular member **171** has a thickness larger than that of each of the discs **161** to **170** and has a higher rigidity than the discs **161** to **170**.

(86) The first passage **75** and the third passage **106** shown in FIG. 2 form a flow path through which the oil fluid L flows from the first chamber **19** toward the second chamber **20** during the extension stroke. The third passage **106** is in constant communication with the first chamber **19**.

(87) As shown in FIG. 4, the discs **162** to **164** are placed on the first seat **46** to form a first damping valve **191** that opens and closes the first passage **75** by being separated from and coming into contact with the first seat **46**. The fixed orifice **184** and the check valve **185** are provided in the first damping valve **191**.

(88) The discs **166** to **168** are placed on the second seat **47** to form a second damping valve **192** that opens and closes the first passage **75** by being separated from and coming into contact with the second seat **47**. The fixed orifice **187** is provided in the second damping valve **192**. The second damping valve **192** is provided on the second chamber **20** side with respect to the first damping valve **191**. A total thickness of the discs **166** to **168** of the second damping valve **192** is larger than a total thickness of the discs **162** to **164** of the first damping valve **191**. The second damping valve **192** has a higher rigidity than the first damping valve **191**.

(89) When the first damping valve **191** is separated from the first seat **46** and opens, the first damping valve **191** allows the oil fluid L to flow from the first chamber **19** shown in FIG. 2 to the second chamber **20** side through the third passage **106** and all the passages forming the first passage **75** including the passages in the plurality of passage holes **52**, the passage in the passage groove **51**, and a passage between the first damping valve **191** and the first seat **46**. At that time, the first damping valve **191** suppresses a flow of the oil fluid L to generate a damping force. The first damping valve **191** is provided in the first passage **75** to generate a damping force by suppressing

the flow of the oil fluid L that occurs during the extension stroke.

(90) Since the second damping valve **192** has a higher rigidity than the first damping valve **191**, the second damping valve **192** is separated from the second seat **47** to open after the first damping valve **191** is separated from the first seat **46** and opened. When the second damping valve **192** is separated from the second seat **47** and opens, the second damping valve **192** allows the oil fluid L to flow from the first chamber **19** to the second chamber **20** side through the third passage **106** and all the passages forming the first passage **75** including the passages in the plurality of passage holes **52**, the passage in the passage groove **51**, the passage between the first damping valve **191** and the first seat **46**, a passage between the first seat **46** and the second seat **47**, and a passage between the second damping valve **192** and the second seat **47**. At that time, the second damping valve **192** suppresses a flow of the oil fluid L to generate a damping force. The second damping valve **192** is provided in the first passage **75** to generate a damping force by suppressing the flow of the oil fluid L that occurs during the extension stroke.

(91) The first damping valve **191** is provided on an upstream side of the second damping valve **192** in a flow direction of the oil fluid L in the extension stroke. The first damping valve **191** has a lower rigidity and a lower valve opening pressure than the second damping valve **192**. Therefore, the first damping valve **191** opens in a region in which a moving speed of the piston **18** (hereinafter referred to as a piston speed) is lower than that of the second damping valve **192**.

(92) As described above, the first piston body **36** includes the first seat **46** and the second seat **47**. The first seat **46** communicates with the first passage **75**, and on which the first damping valve **191** that opens at a low piston speed is placed. The second seat **47** has a larger diameter than the first seat **46**, is provided on the second chamber **20** side with respect to the first seat **46**, and on which the second damping valve **192** that opens after the first damping valve **191** opens is placed.

(93) The check valve **185** provided in the first damping valve **191** closes for a flow in a direction from the first chamber **19** to the second chamber **20** during the extension stroke.

(94) As shown in FIG. 5, one disc **201**, a plurality of (specifically, two) discs **202**, one disc **203**, one disc **204**, a plurality of (specifically, three) discs **205**, a plurality of (specifically, two) discs **206**, one disc **207**, one disc **208**, and one annular member **209** are stacked on the inner seat **85** of the piston **18** in order from the inner seat **85** side in the axial direction of the piston **18**. The number of discs shown here is one embodiment and can be appropriately adjusted according to desired damping force characteristics.

(95) The discs **201** to **208** and the annular member **209** are all made of a metal. The discs **201** to **208** are all formed in a bored circular flat plate shape each having a certain thickness. The annular member **209** has an annular shape. The fitting shaft part **32** of the piston rod **21** is fitted to an inner side of all the discs **201** to **208** and the annular member **209**.

(96) The disc **201** has an outer diameter larger than an outer diameter of the third seat **86** of the piston **18** and smaller than an inner diameter of the fourth seat **87** of the piston **18**. The disc **201** is separated from and comes in contact with the third seat **86** to open and close the fourth passage **107** at a position of the third seat **86**. A notch part **221** is formed in an outer circumferential portion of the disc **201**. The notch part **221** penetrates the disc **201** in the axial direction of the disc **201**. The notch part **221** opens to the outer circumferential portion of the disc **201**. A plurality of notch parts **221** are formed in the disc **201** at regular intervals in the circumferential direction of the disc **201**. The plurality of notch parts **221** extend to cross the third seat **86** in the radial direction. The plurality of notch parts **221** form a fixed orifice **222** in the fourth passage **107** with the disc **201** in contact with the third seat **86**.

(97) The plurality of discs **202** have the same outer diameter, and outer diameters thereof are each the same as the outer diameter of the disc **201**. Of the plurality of discs **202**, the disc **202** on a side closest to the disc **201** covers the plurality of notch parts **221** of the disc **201** from a side opposite to the third seat **86** to form the fixed orifice **222**.

(98) The disc **203** has an outer diameter smaller than the outer diameter of the disc **202** and slightly

larger than an outer diameter of the inner seat **85**.

(99) The disc **204** has an outer diameter larger than an outer diameter of the fourth seat **87** of the piston **18**. The disc **204** is separated from and comes in contact with the fourth seat **87** to open and close the fourth passage **107** at a position of the fourth seat **87**. A notch part **225** is formed in an outer circumferential portion of the disc **204**. The notch part **225** penetrates the disc **204** in the axial direction of the disc **204**. The notch part **225** opens to the outer circumferential portion of the disc **204**. A plurality of notch parts **225** are formed in the disc **204** at regular intervals in the circumferential direction of the disc **204**. The plurality of notch parts **225** extend to cross the fourth seat **87** in the radial direction. The plurality of notch parts **225** form a fixed orifice **226** in the fourth passage **107** with the disc **204** in contact with the fourth seat **87**. A flow path cross-sectional area of the fixed orifice **226** is larger than a flow path cross-sectional area of the fixed orifice **222**.

(100) The plurality of discs **205** have the same outer diameter, and outer diameters thereof are each equal to the outer diameter of the disc **204**. Of the plurality of discs **205**, the disc **205** on a side closest to the disc **204** covers the plurality of notch parts **225** of the disc **204** from a side opposite to the fourth seat **87** to form the fixed orifice **226**.

(101) The plurality of discs **206** have the same outer diameter, and outer diameters thereof are each smaller than the outer diameter of the disc **205** and larger than the outer diameter of the disc **203**.

(102) The disc **207** has an outer diameter smaller than the outer diameter of the disc **206** and equal to the outer diameter of the disc **203**.

(103) The disc **208** has an outer diameter larger than the outer diameter of the disc **207** and equal to the outer diameter of the disc **205**.

(104) The annular member **209** has an outer diameter smaller than the outer diameter of the disc **208** and larger than the outer diameter of the disc **207**. The annular member **209** has a thickness larger than that of each of the discs **201** to **208** and has a higher rigidity than the discs **201** to **208**.

(105) As shown in FIG. 2, the fourth passage **107** and the second passage **76** form a flow path through which the oil fluid L flows from the second chamber **20** toward the first chamber **19** mainly during the compression stroke. The second passage **76** is in constant communication with the second chamber **20**.

(106) The discs **201** and **202** shown in FIG. 5 are placed on the third seat **86** to form a third damping valve **231** that opens and closes the fourth passage **107** by being separated from and coming into contact with the third seat **86**. The fixed orifice **222** is provided in the third damping valve **231**.

(107) The discs **204** to **206** are placed on the fourth seat **87** to form a fourth damping valve **232** that opens and closes the fourth passage **107** by being separated from and coming into contact with the fourth seat **87**. The fixed orifice **226** is provided in the fourth damping valve **232**. The fourth damping valve **232** is provided on the first chamber **19** side with respect to the third damping valve **231**. The fourth damping valve **232** has a total thickness of the discs **204** to **206** that is larger than a total thickness of the discs **201** and **202** of the third damping valve **231** and has a higher rigidity than the third damping valve **231**.

(108) When the third damping valve **231** is separated from the third seat **86** and opens, the third damping valve **231** allows the oil fluid L to flow from the second chamber **20** shown in FIG. 2 to the first chamber **19** side through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, and a passage between the third damping valve **231** and the third seat **86**. At that time, the third damping valve **231** suppresses the flow of the oil fluid L to generate a damping force. The third damping valve **231** is provided in the fourth passage **107** to generate a damping force by suppressing the flow of the oil fluid L that occurs during the compression stroke.

(109) Since the fourth damping valve **232** has a higher rigidity than the third damping valve **231**, the fourth damping valve **232** is separated from the fourth seat **87** to open after the third damping valve **231** is separated from the third seat **86** and opened. When the fourth damping valve **232** is

separated from the fourth seat **87** and opens, the fourth damping valve **232** allows the oil fluid **L** to flow from the second chamber **20** to the first chamber **19** side through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, the passage between the third damping valve **231** and the third seat **86**, the passage between the third seat **86** and the fourth seat **87**, and a passage between the fourth damping valve **232** and the fourth seat **87**. At that time, the fourth damping valve **232** suppresses the flow of the oil fluid **L** to generate a damping force. The fourth damping valve **232** is provided in the fourth passage **107** to generate a damping force by suppressing the flow of the oil fluid **L** that occurs during the compression stroke.

(110) The third damping valve **231** is provided on an upstream side of the fourth damping valve **232** in a flow direction of the oil fluid **L** in the compression stroke. The third damping valve **231** has a lower rigidity and a lower valve opening pressure than the fourth damping valve **232**.

Therefore, the third damping valve **231** opens in a region in which the moving speed of the piston **18** is lower than that of the fourth damping valve **232**.

(111) As shown in FIG. **4**, the check valve **185** provided in the first damping valve **191** opens for a flow of the oil fluid **L** from the second chamber **20** toward the first chamber **19** shown in FIG. **2** during the compression stroke. At this time, the check valve **185** shown in FIG. **4** allows the oil fluid **L** to flow substantially without generating a damping force. The check valve **185** opens for a flow toward the first chamber **19** shown in FIG. **2** and closes for a flow in the opposite direction.

(112) As described above, the second piston body **37** includes the third seat **86** and the fourth seat **87**. The third seat **86** communicates with the fourth passage **107** and, on which the third damping valve **231** that opens at a low piston speed is placed. The fourth seat **87** has a larger diameter than the third seat **86**, is provided on the first chamber **19** side with respect to the third seat **86**, and on which the fourth damping valve **232** that opens after the third damping valve **231** opens is placed.

(113) The first piston body **36** includes the first passage **75** and the second passage **76** that allow the oil fluid **L** to flow between the first chamber **19** on one side and the second chamber **20** on the other side in the cylinder **2** due to movement of the piston **18**. The second piston body **37** includes the third passage **106** and the fourth passage **107** that allow the oil fluid **L** to flow between the first chamber **19** on one side and the second chamber **20** on the other side in the cylinder **2** due to movement of the piston **18**.

(114) In the piston rod **21**, the annular member **209** is placed on the shaft step part **29** with the mounting shaft part **28** inserted through the inside. Then, as shown in FIG. **5**, the disc **208**, the disc **207**, the plurality of discs **206**, the plurality of discs **205**, the disc **204**, the disc **203**, the plurality of discs **202**, the disc **201**, and the piston **18** are stacked in that order on the annular member **209** with the mounting shaft part **28** inserted through the inside of them. In addition, as shown in FIG. **4**, the disc **161**, the disc **162**, the disc **163**, the disc **164**, the disc **165**, the disc **166**, the plurality of discs **167**, the plurality of discs **168**, the disc **169**, the disc **170**, and the annular member **171** are stacked in that order on the piston **18** with the mounting shaft part **28** inserted through the inside of them. In this state, the parts from the annular member **209** to the annular member **171** are fitted to the fitting shaft part **32** of the mounting shaft part **28**. In this state, as shown in FIG. **2**, a nut **241** is screwed onto the screw shaft part **31** of the mounting shaft part **28** that protrudes from the annular member **171**. As a result, inner circumferential sides of the parts from the annular member **209** to the annular member **171** are clamped together in the axial direction by the shaft step part **29** of the piston rod **21** and the nut **241**.

(115) As shown in FIG. **1**, the base member **26** of the base valve **25** is provided between the bottom part **12** of the outer tube **4** and the inner tube **3**. The base valve **25** includes a disc valve **252**, a disc valve **253** and a mounting pin **254** in addition to the base member **26**. The disc valve **252** is provided on a lower side of the base member **26**, that is, on the reservoir chamber **6** side. The disc valve **253** is provided on an upper side of the base member **26**, that is, on the second chamber **20** side. The mounting pin **254** attaches the disc valve **252** and the disc valve **253** to base member **26**.

(116) The base member **26** has an annular shape, and the mounting pin **254** is inserted through a center in the radial direction. A plurality of passage holes **255** and a plurality of passage holes **256** are formed in the base member **26**. The plurality of passage holes **255** allow the oil fluid L to flow between the second chamber **20** and the reservoir chamber **6**. The plurality of passage holes **256** are on an outer side of the plurality of passage holes **255** in the radial direction of the base member **26**. The plurality of passage holes **256** allow the oil fluid L to flow between the second chamber **20** and the reservoir chamber **6**. The disc valve **252** on the reservoir chamber **6** side allows a flow of the oil fluid L from the second chamber **20** to the reservoir chamber **6** through the passage holes **255**. On the other hand, the disc valve **252** suppresses a flow of the oil fluid L from the reservoir chamber **6** to the second chamber **20** through the passage holes **255**. The disc valve **253** allows a flow of the oil fluid L from the reservoir chamber **6** to the second chamber **20** through the passage holes **256**. On the other hand, the disc valve **253** suppresses a flow of the oil fluid L from the second chamber **20** to the reservoir chamber **6** through the passage holes **256**.

(117) The disc valve **252**, together with the base member **26**, forms a damping valve mechanism **257**. The damping valve mechanism **257** opens during the compression stroke of the shock absorber **1** to allow the oil fluid L to flow from the second chamber **20** to the reservoir chamber **6** and generate a damping force. The disc valve **253**, together with the base member **26**, forms a suction valve mechanism **258**. The suction valve mechanism **258** opens during the extension stroke of the shock absorber **1** to allow the oil fluid L to flow from the reservoir chamber **6** into the second chamber **20**. Further, the suction valve mechanism **258** performs a function of allowing the oil fluid L to flow from the reservoir chamber **6** to the second chamber **20** substantially without generating a damping force to supplement a shortage of the liquid caused mainly by extension of the piston rod **21** from the cylinder **2**.

(118) Next, a main operation of the shock absorber **1** will be explained.

(119) "Extension Stroke"

(120) {First Region in which Piston Speed is Lower than First Predetermined Value}

(121) In a first region of the extension stroke, the oil fluid L from the first chamber **19** does not flow to the second chamber **20** through the third passage **106** and the first passage **75** because the check valve **185** of the first damping valve **191** is closed. Therefore, the oil fluid L from the first chamber **19** flows to the second chamber **20** through all the passages forming the fourth passage **107** including the fixed orifice **226** of the fourth damping valve **232**, a passage between the third seat **86** and the fourth seat **87**, the fixed orifice **222** of the third damping valve **231**, the passage in the passage groove **91**, and the passages in the plurality of passage holes **92**, and furthermore through the second passage **76**. Therefore, in the first region of the extension stroke, a damping force having orifice characteristics (in which the damping force is substantially proportional to the square of the piston speed) due to a flow path cross-sectional area a of the fixed orifice **222** is generated.

(122) {Second Region in which Piston Speed is Equal to or Higher than First Predetermined Value and Lower than Second Predetermined Value Larger than First Predetermined Value}

(123) In a second region of the extension stroke, the oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through the third passage **106** and all the passages forming the first passage **75** including the passages in the plurality of passage holes **52**, the passage in the passage groove **51**, the passage between the first damping valve **191** that opens and the first seat **46**, the passage between the first seat **46** and the second seat **47**, and the fixed orifice **187** of the second damping valve **192**. The oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through all the passages forming the fourth passage **107** including the fixed orifice **226** of the fourth damping valve **232**, the passage between the third seat **86** and the fourth seat **87**, the fixed orifice **222** of the third damping valve **231**, the passage in the passage groove **91**, and the passages in the plurality of passage holes **92**, and furthermore through the second passage **76**. Therefore, in the second region of the extension stroke, a damping force having

characteristics combining valve characteristics (in which the damping force is substantially proportional to the piston speed) due to the first damping valve **191** and the orifice characteristics (in which the damping force is substantially proportional to the square of the piston speed) due to the flow path cross-sectional area a of the fixed orifice **222** is generated.

(124) {Third Region in which Piston Speed is Equal to or Higher than Second Predetermined Value and Lower than Third Predetermined Value Larger than Second Predetermined Value}

(125) In a third region of the extension stroke, the oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through the third passage **106** and all the passages forming the first passage **75** including the passages in the plurality of passage holes **52**, the passage in the passage groove **51**, the passage between the first damping valve **191** that opens and the first seat **46**, the passage between the first seat **46** and the second seat **47**, and the fixed orifice **187** of the second damping valve **192**. At that time, the flow of the oil fluid L is reduced by the fixed orifice **187**. The oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through all the passages forming the fourth passage **107** including the fixed orifice **226** of the fourth damping valve **232**, the passage between the third seat **86** and the fourth seat **87**, the fixed orifice **222** of the third damping valve **231**, the passage in the passage groove **91**, and the passages in the plurality of passage holes **92**, and furthermore through the second passage **76**. Therefore, in the third region of the extension stroke, a damping force having characteristics combining orifice characteristic due to flow path cross-sectional area B of the fixed orifice **187** and the orifice characteristics due to the flow path cross-sectional area a of the fixed orifice **222** is generated.

(126) {Fourth Region in which Piston Speed is Equal to or Higher than Third Predetermined Value}

(127) In a fourth region of the extension stroke, the oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through the third passage **106** and all the passages forming the first passage **75** including the passages in the plurality of passage holes **52**, the passage in the passage groove **51**, the passage between the first damping valve **191** that opens and the first seat **46**, the passage between the first seat **46** and the second seat **47**, and the passage between the second damping valve **192** that opens and the second seat **47**. The oil fluid L from the first chamber **19**, on the other hand, flows to the second chamber **20** through all the passages forming the fourth passage **107** including the fixed orifice **226** of the fourth damping valve **232**, the passage between the third seat **86** and the fourth seat **87**, the fixed orifice **222** of the third damping valve **231**, the passage in the passage groove **91**, and the passages in the plurality of passage holes **92**, and furthermore through the second passage **76**. Therefore, in the fourth region of the extension stroke, a damping force having characteristics combining valve characteristic due to the second damping valve **192** and the orifice characteristics due to the flow path cross-sectional area a of the fixed orifice **222** is generated.

(128) "Compression Stroke"

(129) {Fifth Region in which Piston Speed is Lower than Fifth Predetermined Value}

(130) In a fifth region of the compression stroke, the oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, the fixed orifice **222** of the third damping valve **231**, the passage between the third seat **86** and the fourth seat **87**, and the fixed orifice **226** of the fourth damping valve **232**. The oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through all the passages forming the first passage **75** including the fixed orifice **187** of the second damping valve **192**, the passage between the first seat **46** and the second seat **47**, the fixed orifice **184** of the first damping valve **191**, the check valve **185** that opens, the passage in the passage groove **51**, and the passages in the plurality of passage holes **52**, and through the third passage **106**. Therefore, in the fifth region of the compression stroke, a damping force having

characteristics combining the orifice characteristics due to the flow path cross-sectional area a of the fixed orifice **222** of the third damping valve **231** and orifice characteristics due to a flow path cross-sectional area b of the fixed orifice **184** of the first damping valve **191** is generated.

(131) {Sixth Region in which Piston Speed is Equal to or Higher than Fifth Predetermined Value and Lower than Sixth Predetermined Value Larger than Fifth Predetermined Value}

(132) In a sixth region of the compression stroke, the oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, the passage between the third damping valve **231** that opens and the third seat **86**, the passage between the third seat **86** and the fourth seat **87**, and the fixed orifice **226** of the fourth damping valve **232**. The oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through all the passages forming the first passage **75** including the fixed orifice **187** of the second damping valve **192**, the passage between the first seat **46** and the second seat **47**, the fixed orifice **184** of the first damping valve **191**, the check valve **185** that opens, the passage in the passage groove **51**, and the passages in the plurality of passage holes **52**, and through the third passage **106**. Therefore, in the sixth region of the compression stroke, a damping force having characteristics combining valve characteristics due to the third damping valve **231** and the orifice characteristics due to the flow path cross-sectional area b of the fixed orifice **184** of the first damping valve **191** is generated.

(133) {Seventh Region in which Piston Speed is Equal to or Higher than Sixth Predetermined Value and Lower than Seventh Predetermined Value Larger than Sixth Predetermined Value}

(134) In a seventh region of the compression stroke, the oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, the passage between open third damping valve **231** that opens and the third seat **86**, the passage between the third seat **86** and the fourth seat **87**, and the fixed orifice **226** of the fourth damping valve **232**. At that time, the flow of the oil fluid L is reduced by the fixed orifice **226**. The oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through all the passages forming the first passage **75** including the fixed orifice **187** of the second damping valve **192**, the passage between the first seat **46** and the second seat **47**, the fixed orifice **184** of the first damping valve **191**, the check valve **185** that opens, the passage in the passage groove **51**, and the passages in the plurality of passage holes **52**, and through the third passage **106**. Therefore, in the seventh region of the compression stroke, a damping force having characteristics combining orifice characteristics due to a flow path cross-sectional area A of the fixed orifice **226** of the fourth damping valve **232** and the orifice characteristics due to the flow path cross-sectional area b of the fixed orifice **184** of the first damping valve **191** is generated.

(135) {Eighth Region in which Piston Speed is Equal to or Higher than Seventh Predetermined Value}

(136) In an eighth region of the compression stroke, the oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through the second passage **76** and all the passages forming the fourth passage **107** including the passages in the plurality of passage holes **92**, the passage in the passage groove **91**, the passage between the third damping valve **231** that opens and the third seat **86**, the passage between the third seat **86** and the fourth seat **87**, and the passage between the fourth damping valve **232** that opens and the fourth seat **87**. The oil fluid L from the second chamber **20**, on the other hand, flows to the first chamber **19** through all the passages forming the first passage **75** including the fixed orifice **187** of the second damping valve **192**, the passage between the first seat **46** and the second seat **47**, the fixed orifice **184** of the first damping valve **191**, the check valve **185** that opens, the passage in the passage groove **51**, and the passages in the plurality of passage holes **52**, and through the third passage **106**. Therefore, in the seventh region of the compression stroke, a damping force having characteristics combining valve

characteristics due to the fourth damping valve **232** and the orifice characteristics due to the flow path cross-sectional area **b** of the fixed orifice **184** of the first damping valve **191** is generated. (137) In the shock absorber **1**, the check valve **185** is provided in the first damping valve **191** which is one of the first damping valve **191** and the third damping valve **231**, which both open at a low piston speed, and a check valve is not provided in the third damping valve **231** which is the other thereof. Therefore, as described above, in the extension stroke, orifice characteristics before the first damping valve **191** opens are the characteristics only due to the flow path cross-sectional area **a** of the fixed orifice **222** of the third damping valve **231**, and orifice characteristics after the first damping valve **191** opens are the characteristics due to the flow path cross-sectional area **a** of the fixed orifice **222** of the third damping valve **231** and the flow path cross-sectional area **B** of the fixed orifice **187** of the second damping valve **192**. On the other hand, in the compression stroke, orifice characteristics before the third damping valve **231** opens are the characteristics due to the flow path cross-sectional area **a** of the fixed orifice **222** of the third damping valve **231** and the flow path cross-sectional area **b** of the fixed orifice **184** of the first damping valve **191**, and orifice characteristics after the third damping valve **231** opens are the characteristics due to the flow path cross-sectional area **A** of the fixed orifice **226** of the fourth damping valve **232** and the flow path cross-sectional area **b** of the fixed orifice **184** of the first damping valve **191**.

(138) Further, in the above-described description, a case in which the fixed orifice **184** and the check valve **185** are provided in the first damping valve **191** has been described as an example. Alternatively, similar fixed orifice and a check valve may be provided in the third damping valve **231**. In that case, for example, the entirety of the first damping valve **191** including the check valve **185**, the piston **18**, and the third damping valve **231**, as it is, is reversed in the axial direction with respect to the above-described configuration and fitted to the fitting shaft part **32**. Then, the reversed first damping valve **191** including the fixed orifice **184** and the check valve **185** serves as a third damping valve including a third fixed orifice and a third check valve, and the reversed third damping valve **231** serves as a first damping valve. The third check valve provided in the third damping valve in this manner allows a flow of the oil fluid **L** from the first chamber **19** to the second chamber **20** through the third fixed orifice, the fourth passage **107**, and the second passage **76**. Also, the third check valve restricts a flow of the oil fluid **L** from the second chamber **20** to the first chamber **19** through the second passage **76**, the fourth passage **107**, and the third fixed orifice.

(139) Also, a configuration in which a second fixed orifice and a second check valve that allows the oil fluid **L** to flow in only one direction to the second fixed orifice are formed of laminated discs may be provided in the second damping valve **192**. In this case, the second check valve allows a flow of the oil fluid **L** from the second chamber **20** to the first chamber **19** through the second fixed orifice, the first passage **75**, and the third passage **106**. Also, the second check valve restricts a flow of the oil fluid **L** from the first chamber **19** to the second chamber **20** through the third passage **106**, the first passage **75**, and the second fixed orifice. In that case, only a fixed orifice may be provided in the first damping valve **191** without providing the check valve **185**. Also, both the first damping valve **191** including the fixed orifice **184** and the check valve **185** and the second damping valve **192** including the second fixed orifice and the second check valve may be provided.

(140) Also, a configuration in which a fourth fixed orifice and a fourth check valve that allows the oil fluid **L** to flow in only one direction to the fourth fixed orifice are formed of laminated discs may be provided in the fourth damping valve **232**. In this case, the fourth check valve allows a flow of the oil fluid **L** from the first chamber **19** to the second chamber **20** through the fourth fixed orifice, the fourth passage **107**, and the second passage **76**. Also, the fourth check valve restricts a flow of the oil fluid **L** from the second chamber **20** to the first chamber **19** through the second passage **76**, the fourth passage **107**, and the fourth fixed orifice. Both the third damping valve **231** including the third fixed orifice and the third check valve and the fourth damping valve **232** including the fourth fixed orifice and the fourth check valve may be provided.

(141) Japanese Unexamined Patent Application, First Publication No. 2010-107003, Japanese

Unexamined Patent Application, First Publication No. 2014-129865, Japanese Unexamined Patent Application, First Publication No. H2-113142, and the specification of United States Patent Application, Publication No. 2013/0037361 described above disclose a shock absorber in which a piston is constituted by a plurality of piston bodies. Also, the specification of United States Patent Application, Publication No. 2013/0037361 and Japanese Unexamined Utility Model Application, First Publication No. H1-149037 described above disclose a shock absorber including a valve and a valve that opens in a region in which a piston speed is higher than that of the above-described valve. Incidentally, in shock absorbers, there has been a demand to minimize an increase in the number of parts to suppress a decrease in productivity. For example, a shock absorber including a first damping valve and a second damping valve that opens in a region in which a piston speed is higher than that of the first damping valve on an extension side, and a third damping valve and a fourth damping valve that opens in a region in which a piston speed is higher than that of the third damping valve on a compression side may be configured. In this case, since the number of valves is large, it is likely to be accompanied by a decrease in productivity. For this, it is desirable to minimize an increase in the number of parts to suppress a decrease in productivity.

(142) In the shock absorber **1** of the embodiment, the piston **18** includes the first piston body **36**, the second piston body **37**, and the piston band **38**. The first piston body **36** includes the first passage **75**, the second passage **76**, the first seat **46**, and the second seat **47**. The first passage **75** and the second passage **76** allow the oil fluid L serving as a working fluid to flow between the first chamber **19** on one side and the second chamber **20** on the other side in the cylinder **2** due to movement of the piston **18**. The first seat **46** communicates with the first passage **75**, and on which the first damping valve **191** that opens at a low piston speed is placed. The second seat **47** has a larger diameter than the first seat **46**, is provided on the second chamber **20** side, and on which the second damping valve **192** that opens after the first damping valve **191** opens is placed. The second piston body **37** includes the third passage **106**, the fourth passage **107**, the third seat **86**, and the fourth seat **87**. The third passage **106** and the fourth passage **107** allow the oil fluid L serving as a working fluid to flow between the first chamber **19** on one side and the second chamber **20** on the other side in the cylinder **2** due to movement of the piston **18**. The third seat **86** communicates with the fourth passage **107**, and on which the third damping valve **231** that opens at a low piston speed is placed. The fourth seat **87** has a larger diameter than the third seat **86**, is provided on the first chamber **19** side, and on which the fourth damping valve **232** that opens after the third damping valve **231** opens is placed. Since the piston **18** includes the first piston body **36**, the second piston body **37**, and the piston band **38**, the number of components of the piston **18** can be reduced.

Therefore, man-hours for manufacturing each part constituting the piston **18** and man-hours for the assembly can be reduced. Also, the piston **18** can be assembled as a sub-assembly before being assembled to the piston rod **21**. Therefore, man-hours for assembling parts to the piston rod **21** can be reduced. Therefore, a decrease in productivity can be suppressed. The shock absorber **1** has a structure in which the first damping valve **191**, the second damping valve **192**, the third damping valve **231**, and the fourth damping valve **232** are placed on the piston **18**, and this is a structure that is particularly likely to be accompanied by a decrease in productivity. Therefore, suppressing a decrease in productivity as described above greatly contributes to an effect of minimizing an increase in product costs of the shock absorber **1**.

(143) Also, the shock absorber **1** includes the positioning part **121** provided on the joint surfaces **41** and **81** of the first piston body **36** and the second piston body **37** that are connected to each other. The positioning part **121** aligns positions of the first passage **75** and the third passage **106** and aligns positions of the second passage **76** and the fourth passage **107**. Therefore, positioning of these can be easily performed. Also, in the shock absorber **1**, since an inner diameter of the first piston body **36** and an inner diameter of the second piston body **37** are different, when one of the first piston body **36** and the second piston body **37** having a smaller inner diameter is fitted to the piston rod **21**, the piston **18** can be positioned on the piston rod **21**. Therefore, when the first piston

body **36** and the second piston body **37** are connected, the inner circumferential surfaces are not necessary to be aligned with each other. Therefore, a decrease in productivity can be further suppressed.

(144) Also, the shock absorber **1** includes the check valve **185** that opens for a flow toward the first chamber **19** or the second chamber **20** and closes for a flow in the opposite direction in the first damping valve **191** or the third damping valve **231**. Therefore, orifice characteristics before the first damping valve **191** or the third damping valve **231** opens can be made different between the extension stroke and the compression stroke. For example, when the check valve **185** is provided in the first damping valve **191** as described above, in the extension stroke, before the first damping valve **191** opens, orifice characteristics are due to the flow path cross-sectional area *a* of the fixed orifice **222**, and after the first damping valve **191** opens, orifice characteristics are due to the flow path cross-sectional area *a* of the fixed orifice **222** and the flow path cross-sectional area *B* of the fixed orifice **187**. In the compression stroke, before the third damping valve **231** opens, orifice characteristics are due to the flow path cross-sectional area *a* of the fixed orifice **222** and the flow path cross-sectional area *b* of the fixed orifice **184**, and after the third damping valve **231** opens, orifice characteristics are due to the flow path cross-sectional area *A* of the fixed orifice **226** and the flow path cross-sectional area *b* of the fixed orifice **184**.

(145) A manufacturing method of the shock absorber **1** of the embodiment includes first to third steps. In the first step, the joint surface **41** of the first piston body **36** and the joint surface **81** of the second piston body **37** are connected so that the first passage **75** and the third passage **106** communicate, and the second passage **76** and the fourth passage **107** communicate. In the second step, an end portion of the first passage **75** opposite to the third passage **106**, an end portion of the third passage **106** opposite to the first passage **75**, an end portion of the second passage **76** opposite to the fourth passage **107**, an end portion of the fourth passage **107** opposite to the second passage **76**, and the deep groove part **131** and the shallow groove parts **132** to which the piston band **38** is attached are processed. In the third step, the piston band **38** is attached to the deep groove part **131** and the shallow groove parts **132**. In this way, since the first piston body **36** and the second piston body **37** are processed in the second step while the first piston body **36** and the second piston body **37** are connected in the first step, relative accuracies between parts can be easily secured. Also, when the first to third steps are performed, the piston **18** can be assembled as a sub-assembly before being assembled to the piston rod **21**. Therefore, man-hours for assembling parts to the piston rod **21** can be reduced. Therefore, a decrease in productivity can be suppressed.

(146) In the third step, the piston band **38** is attached to the deep groove part **131** via the second seat **47**, and then is attached to the plurality of shallow groove parts **132** having a depth smaller than that of the deep groove part **131**. Therefore, the piston band **38** can be properly and smoothly attached to the first piston body **36** and the second piston body **37**.

(147) Further, in the embodiment, a hydraulic shock absorber has been shown as an example of the shock absorber **1**, but the above-described structure can also be employed for a shock absorber using water or air as a working fluid.

EXPLANATION OF REFERENCES

(148) **1** Shock absorber **2** Cylinder **18** Piston **19** First chamber (chamber) **20** Second chamber (chamber) **21** Piston rod **36** First piston body **37** Second piston body **38** Piston band **41**, **81** Joint surface **46** First seat **47** Second seat **75** First passage **76** Second passage **86** Third seat **87** Fourth seat **106** Third passage **107** Fourth passage **121** Positioning part **185** Check valve **191** First damping valve **192** Second damping valve **231** Third damping valve **232** Fourth damping valve

Claims

1. A shock absorber comprising: a cylinder in which a working fluid is sealed; a piston fitted in the cylinder to be slidable and partitioning an inside of the cylinder into two chambers; and a piston

rod connected to the piston and extending to an outside of the cylinder, wherein the piston includes: a first piston body having a first passage and a second passage configured to allow the working fluid to flow between the two chambers in the cylinder due to movement of the piston, a first seat communicating with the first passage and on which a first damping valve which opens at a low piston speed is placed, and a second seat having a larger diameter than the first seat, provided on a side of one of the two chambers, and on which a second damping valve which opens after the first damping valve opens is placed; and a second piston body having a third passage and a fourth passage configured to allow the working fluid to flow between the two chambers in the cylinder due to movement of the piston, a third seat communicating with the fourth passage and on which a third damping valve which opens at a low piston speed is placed, and a fourth seat having a larger diameter than the third seat, provided on a side of another one of the two chambers, and on which a fourth damping valve which opens after the third damping valve opens is placed, the first piston body and the second piston body are connected so that the first passage communicates with the third passage and the second passage communicates with the fourth passage, an inner diameter of the first piston body and an inner diameter of the second piston body are different, and one of the first piston body and the second piston body having a smaller inner diameter is in contact with and attached to the piston rod.

2. The shock absorber according to claim 1, wherein a check valve which opens for a flow toward the one of the two chambers or the another one of the two chambers and closes for a flow in an opposite direction is provided in the first damping valve or the third damping valve.

3. A manufacturing method of a shock absorber which is a manufacturing method of a shock absorber including: a cylinder in which a working fluid is sealed; a piston fitted in the cylinder to be slidable and partitioning an inside of the cylinder into two chambers; and a piston rod connected to the piston and extending to an outside of the cylinder, wherein the piston includes: a first piston body having a first passage and a second passage configured to allow the working fluid to flow between the two chambers in the cylinder due to movement of the piston, a first seat communicating with the first passage and on which a first damping valve which opens at a low piston speed is placed, and a second seat having a larger diameter than the first seat, provided on a side of one of the two chambers, and on which a second damping valve which opens after the first damping valve opens is placed; a second piston body having a third passage and a fourth passage configured to allow the working fluid to flow between the two chambers in the cylinder due to movement of the piston, a third seat communicating with the fourth passage and on which a third damping valve which opens at a low piston speed is placed, and a fourth seat having a larger diameter than the third seat, provided on a side of another one of the two chambers, and on which a fourth damping valve which opens after the third damping valve opens is placed; and a piston band provided in the first piston body and the second piston body, the manufacturing method comprising: a first step of connecting one surface of the first piston body and one surface of the second piston body so that the first passage communicates with the third passage and the second passage communicates with the fourth passage; a second step of processing an end portion of the first passage opposite to the third passage, an end portion of the third passage opposite to the first passage, an end portion of the second passage opposite to the fourth passage, an end portion of the fourth passage opposite to the second passage, and a groove to which the piston band is attached; and a third step of attaching the piston band to the groove.

4. The manufacturing method of a shock absorber according to claim 3, wherein the groove includes: a plurality of shallow groove parts; and a deep groove part provided on the second seat side with respect to the plurality of shallow groove parts and having a depth larger than that of the plurality of shallow groove parts, and the third step is configured to attach the piston band to the deep groove part via the second seat and then attach the piston band to the plurality of shallow groove parts.
